FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Postpartum Haemorrhage

Tutorial | Week 3 · Medical disorders & PPH | Obstetrics | Dr Kavuma

Primary PPH is blood loss of 500 mL or more within 24 hours of birth (or 300 mL or more with abnormal vital signs, WHO 2025); it is the leading killer of Ugandan mothers, and survival depends on measuring blood loss objectively and starting the E-MOTIVE bundle within minutes, not on estimating by eye.

Must know before the ward round

1. Primary PPH: ≥500 mL within 24 h of birth; WHO/FIGO/ICM 2025 also say act at ≥300 mL plus any abnormal vital sign. Severe/major PPH: ≥1000 mL. Secondary PPH: 24 h to 12 weeks.
2. Causes = the 4 Ts: Tone (about 70%), Trauma (about 20%), Tissue (about 10%), Thrombin (under 1%). Always check all four.
3. Prevention (AMTSL): oxytocin 10 IU IM/IV within 1 minute of birth, delayed cord clamping, controlled cord traction, then check uterine tone every 15 minutes for 2 hours.
4. E-MOTIVE bundle, started together as soon as PPH is diagnosed: uterine Massage, Oxytocics, Tranexamic acid, IV fluids, Vaginal/genital-tract Examination, then Escalate if bleeding continues.
5. Treatment oxytocin: 10 IU IV slowly (or IM), then 20 IU in 1 L NS/RL at 60 drops/min (maintenance 40 drops/min); no more than 3 L of oxytocin-containing fluid.
6. Tranexamic acid 1 g IV over 10 minutes within 3 hours of birth; a second 1 g if bleeding continues after 30 minutes or restarts within 24 hours (WOMAN trial). No benefit after 3 hours.
7. Second-line uterotonics: misoprostol 800 µg sublingual; ergometrine 0.2 mg IM (max 1 mg/24 h; not in hypertension or heart disease); carboprost 0.25 mg deep IM (max 2 mg; not in asthma).
8. Calibrated drape: E-MOTIVE (NEJM 2023) cut severe PPH, laparotomy or death from bleeding by 60% (1.6% vs 4.3%).
9. Visual estimation underestimates blood loss by 30–50%; weigh pads (1 g ≈ 1 mL). Shock index (HR ÷ SBP) ≥0.9 is abnormal and ≥1.4 needs urgent action; hypotension is a late sign.
10. Firm uterus but still bleeding = trauma (inspect cervix and vagina in good light) or thrombin (bedside clotting test: no clot by 7 min), not a reason for another uterotonic.
11. Retained placenta: not out 30 minutes after birth → empty the bladder, extra oxytocin 10 IU, repeat CCT, then manual removal with a single-dose antibiotic. Never ergometrine.
12. Shock out of proportion to visible loss + fundus not palpable = uterine inversion: stop oxytocin, replace immediately, then give oxytocin.

1 Definition
2 Epidemiology
3 Aetiology and Risk Factors
4 Pathophysiology
5 Classification / Staging
6 Clinical Features
7 Differential Diagnoses
8 Clinical Workup
9 Management
10 Complications
11 Prognosis
12 Prevention
13 Special Populations
14 Key Points
15 References

1. Definition

Postpartum haemorrhage (PPH) is excessive bleeding from the genital tract after birth. It is the **number-one direct cause of maternal death in Uganda** and worldwide. Most of these deaths happen **within 24 hours of birth, often within the first 2 hours**, and **most are preventable**.

Term	Definition
Primary PPH	Blood loss **≥500 mL** from the genital tract within **24 hours** of birth (any route). The WHO/FIGO/ICM **2025** guideline adds: diagnose PPH at **≥300 mL plus any abnormal vital sign** (e.g. tachycardia, hypotension)
Severe / major PPH	**≥1000 mL** (RCOG: moderate 1000–2000 mL; severe >2000 mL). Minor = 500–1000 mL
PPH after caesarean	Classically ≥1000 mL; but treat **any bleeding that threatens haemodynamic stability** as PPH
Clinical definition	Any blood loss enough to cause haemodynamic instability. A small, anaemic woman (Hb 8 g/dL) may collapse after "only" 400 mL
Secondary PPH	Abnormal or excessive bleeding from **24 hours to 12 weeks** after birth
Refractory PPH	Bleeding that continues despite first-line uterotonics and the first-response bundle
Retained placenta	Placenta not delivered within **30 minutes** of birth with active management (WHO; UK uses 60 minutes)

The modern approach, in the **WHO 2023 recommendations** and the **WHO/FIGO/ICM 2025 consolidated PPH guideline**, rests on two ideas:

1. **Detect early**: *measure* blood loss (calibrated drape) instead of guessing, and act at **300 mL if vital signs are abnormal**.
2. **Treat as a bundle**: give the **E-MOTIVE** first-response bundle all at once and quickly, not one drug at a time waiting to see what works.

> **CLINICAL PEARL**
> The one thing never to forget: **a woman who bleeds after delivery can die within 2 hours.** Call for help early, measure, massage, give oxytocin **and** tranexamic acid, run fluids, look for the cause (4 Ts), and escalate. Don't wait for her blood pressure to fall.

2. Epidemiology

- Globally PPH affects millions of women every year. It is the single largest cause of maternal death, most of it in sub-Saharan Africa and South Asia.
- **Uganda:** the MMR fell from **336 per 100,000 live births (UDHS 2016) to 189 (UDHS 2022)**. That is still far above the SDG target of <70. **Haemorrhage remains the leading direct cause** of maternal death in MPDSR reports, together with hypertensive disorders, sepsis and abortion complications.
- Ugandan studies report PPH (≥500 mL) in about **9% of births**, with severe PPH (≥1000 mL) in about 1%. That is probably an underestimate, because blood loss is estimated by eye.

UGANDA CONTEXT

The realities on a Ugandan labour ward matter. **Blood** is often unavailable at HC IVs and even at regional referral hospitals (RRHs), where it comes from the Uganda Blood Transfusion Service. **Oxytocin quality** suffers when the cold chain fails (hence interest in **heat-stable carbetocin**). Many women are **anaemic** before they bleed. Night-time deliveries are covered by one midwife. Referral from HC III to HC IV to RRH can take hours. Your management must work with what is on the shelf: oxytocin, misoprostol, ergometrine, TXA, IV fluids, a Foley catheter and a condom, and your hands.

3. Aetiology and Risk Factors

Causes: the 4 Ts

T	Approx. share	Examples
Tone (atony)	~70%	Overdistension (twins, polyhydramnios, macrosomia), prolonged or precipitate labour, prolonged oxytocin augmentation, chorioamnionitis, grand multiparity, fibroids, $MgSO_4$, general anaesthesia, full bladder
Trauma	~20%	Perineal, vaginal and cervical tears, episiotomy extension, uterine rupture, uterine inversion, broad-ligament haematoma, CS extensions
Tissue	~10%	Retained placenta, retained cotyledon or membranes, placenta accreta spectrum (PAS), clots in the cavity
Thrombin	<1%	DIC (abruption, amniotic fluid embolism, sepsis, IUFD, severe pre-eclampsia/HELLP), anticoagulants, inherited bleeding disorders, **dilutional coagulopathy after large transfusion**

Secondary PPH causes: **endometritis** (most common), **retained products**, subinvolution of the placental site, rarely trophoblastic disease (GTN) or arteriovenous malformation.

Risk factors

Two-thirds of PPH happens in women with no risk factor, so prepare for every birth.

Antenatal	Intrapartum
Previous PPH (strongest), grand multiparity (≥5), multiple pregnancy, polyhydramnios, macrosomia, anaemia (Hb <9 g/dL), placenta praevia/PAS, abruption, pre-eclampsia, fibroids, obesity, maternal age >35, previous CS	Prolonged labour (1st or 2nd stage), prolonged oxytocin use, induction, chorioamnionitis/fever, instrumental birth, CS (especially emergency), episiotomy, retained placenta, general anaesthesia, precipitate labour

4. Pathophysiology

Why the placental bed bleeds so much

- At term, **uterine blood flow is 500–800 mL/min** (about 10–15% of cardiac output). The spiral arteries have been converted by trophoblast into **wide, low-resistance, non-muscular channels**. They **cannot constrict themselves**.
- After the placenta separates, bleeding is stopped mainly by **mechanical compression**. The criss-cross (interlacing) fibres of the myometrium contract and kink the vessels running between them. These are the **"living ligatures"**, and they explain why **uterine tone is everything**. Clotting and local thrombosis are secondary.
- **So:** a soft, "boggy" uterus = open vessels = torrential bleeding (tone). A well-contracted uterus that still bleeds must be bleeding from **somewhere else** (trauma or thrombin), or something is preventing contraction

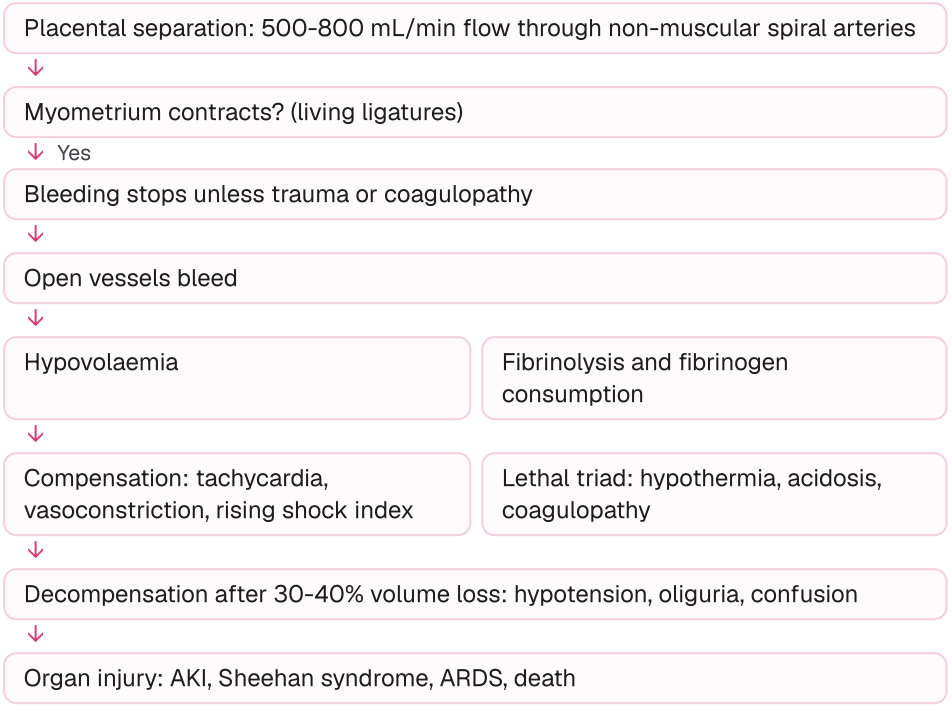
(**tissue**: a retained cotyledon holds the uterus open).

How the body compensates, and why that fools us

- Pregnancy increases blood volume by about **40–50%** (roughly 1.5 L extra). A healthy woman at term can lose **1000 mL with only mild tachycardia**. Young women hold their blood pressure through vasoconstriction until very late.
- **Hypotension is a late sign** (often after 30–40% of blood volume is lost). **Tachycardia, pallor, cool peripheries, restlessness, reduced urine output and a rising shock index** come first.
- **Shock index (SI) = heart rate ÷ systolic BP.** Normal postpartum SI is 0.7–0.9. **SI ≥0.9** should prompt concern and **SI ≥1.4** means urgent action and transfer.

The lethal triad. Continued bleeding leads to **hypothermia** (cold fluids, exposure), **acidosis** (hypoperfusion) and **coagulopathy** (consumption plus dilution with crystalloid). Each worsens the others. Fibrinogen falls first in obstetric haemorrhage, and **fibrinogen <2 g/L predicts progression to severe PPH**. This is why we warm fluids, limit crystalloid, give **tranexamic acid early** (to block fibrinolysis before clots dissolve), and transfuse blood and clotting factors rather than litres of saline.

Why tranexamic acid must be early. TXA blocks plasminogen activation (antifibrinolytic). Fibrinolysis rises sharply within minutes to hours of birth and bleeding. In the **WOMAN trial** (20,060 women), TXA **reduced death from bleeding by about 30% when given within 3 hours**. Given later there was no benefit. Every 15 minutes of delay reduces its benefit.



Feature links

- Atonic myometrium → open spiral arteries → heavy bleeding with clots, soft boggy fundus above the umbilicus.
- Massage → contraction → fundus firms and bleeding slows (confirms tone as the cause).
- Retained cotyledon or clots → uterus cannot stay contracted → fundus that firms then relaxes again and again.
- Tear in cervix or vagina → bleeding independent of tone → bright red trickle with a firm uterus.
- Bleeding into tissue planes → vulval, vaginal or broad-ligament haematoma → severe perineal or rectal pain, unexplained tachycardia, little external blood.

- Uterine rupture → bleeding into the abdomen → abdominal pain, peritonism, easily felt fetal parts, shock greater than visible loss.
- Inverted fundus → traction on peritoneum and ovaries (vagal) → profound shock with bradycardia, fundus not palpable, mass in the vagina.
- Volume loss → sympathetic drive → tachycardia, pallor, sweating, cool peripheries, capillary refill >2 s, rising shock index.
- Reduced renal perfusion → oliguria (<30 mL/h) → acute tubular necrosis if prolonged.
- Reduced cerebral perfusion → restlessness, then confusion and drowsiness (late).
- Fibrinogen consumption and dilution → coagulopathy → no clot at 7 minutes, oozing from cannula sites, the episiotomy and gums.
- Pituitary hypoperfusion (enlarged in pregnancy) → Sheehan syndrome → failed lactation, amenorrhoea.
- Acute whole-blood loss without time for haemodilution → first Hb looks near-normal despite a large bleed.

5. Classification / Staging

By timing: primary (≤24 h) versus secondary (24 h to 12 weeks) (section 1).

By volume: minor 500–1000 mL; major ≥1000 mL (RCOG subdivides into moderate 1000–2000 mL and severe >2000 mL). WHO/FIGO/ICM 2025: PPH is diagnosed at ≥300 mL plus any abnormal vital sign.

By cause: the 4 Ts (section 3).

By haemodynamic compromise: shock index

Shock index (HR ÷ SBP)	Meaning
0.7–0.9	Normal postpartum
≥0.9	Abnormal: concern, look for ongoing bleeding
≥1.4	Urgent: senior at the bedside, blood, theatre or transfer
≥1.7	Critical

Rule of 30 (quick reference): a fall in systolic BP of about 30 mmHg with a rise in pulse of about 30/min suggests loss of roughly **30% of blood volume** (moderate shock). See worked example 1, section 8.

6. Clinical Features

History

In the acute setting, history is taken **while** resuscitation happens, from the woman, the midwife, the partograph and the file. Each item changes the plan.

Question	Why it matters
Time of birth	Still within 3 h? TXA works only then; also tells you whether this is primary or secondary PPH
Mode of birth: SVD, vacuum, CS, fundal pressure	Instrumental = tear; CS = intra-abdominal bleeding; fundal pressure or cord traction on an unseparated placenta = inversion
Was AMTSL given? Which drug, when?	No prophylactic oxytocin makes atony more likely; you also need the total uterotonic dose so far
Placenta out? Complete?	Retained or incomplete placenta = tissue; plan MRP

Question	Why it matters
Labour: length, oxytocin augmentation, fever or chorioamnionitis, precipitate birth	Prolonged or augmented labour and chorioamnionitis → atony; fever → sepsis-related atony and DIC
Baby: weight, twins, polyhydramnios, stillbirth	Overdistension → atony; IUFD or abruption → DIC
Parity and uterine history: grand multipara, previous PPH, previous CS or myomectomy	Atony, rupture; previous CS + praevia → accreta
Blood pressure, pre-eclampsia, heart disease	**No ergometrine**; HELLP → low platelets (thrombin)
Asthma	**No carboprost** (bronchospasm)
Antenatal Hb, blood group, HIV status	Anaemic women decompensate early; plan transfusion; Rh status for anti-D
Known bleeding disorder, anticoagulant, abruption, IUFD	Thrombin cause
Symptoms: dizziness, thirst, breathlessness, confusion	Hypovolaemia; confusion = late, severe shock
Severe perineal or rectal pain	Expanding haematoma
Abdominal pain	Rupture; post-CS bleeding

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted to the postpartum emergency. In an actively bleeding woman the **hand on the fundus and the ABCDE** (section 9) come first; this sequence is completed by the team while the bundle runs.

Introduction, consent and preparation

- Introduce yourself, confirm her identity, explain briefly what you are doing and why, and gain verbal consent (consent for the genital-tract examination and any procedure; a chaperone is present).
- **Hand hygiene**; sterile gloves for the genital-tract examination.
- **Position and exposure**: lie flat (legs raised if shocked); expose the abdomen and perineum; **good light**, an assistant and a speculum or retractors for the genital tract; keep her warm and her dignity covered.

General inspection (end of the bed)

Sign	Mechanism	What it means
Restlessness, air hunger	Hypoperfusion, hypoxia	Significant hypovolaemia
Confusion, drowsiness (AVPU/GCS)	Cerebral hypoperfusion	Late, severe shock
Pallor, sweating	Anaemia; sympathetic drive	Blood loss; low reserve
Blood in the drape, on the bed and floor	Ongoing loss	Read the drape, weigh the linen

Vital signs every 5–15 min: pulse (rate and volume), BP, RR, SpO_2, temperature, **shock index**, **urine output** via catheter (target **≥30 mL/h**).

Hands, face and neck

Sign	Mechanism	What it means
Cold, clammy peripheries; capillary refill >2 s	Peripheral vasoconstriction	Compensated shock
Oozing from cannula sites, gums; petechiae	Consumptive coagulopathy	DIC
Conjunctival pallor	Anaemia	Pre-existing anaemia or acute loss

Abdomen (postpartum)

- **Inspection**: distension (post-CS intra-abdominal bleeding, rupture), scars (previous CS).
- **Palpation of the fundus**: height, consistency (**soft and above the umbilicus** = atony; firm and central = look elsewhere), deviated (full bladder), does it relax again after massage (tissue)?
- Tenderness, guarding, distension, free fluid, easily felt fetal parts or a scar → rupture; post-CS bleeding.
- **Fundus not palpable + shock out of proportion to bleeding → uterine inversion**.

Blood loss: measure it

Read the **calibrated drape**, weigh pads and linen (1 g ≈ 1 mL), count soaked swabs; look under the woman and on the floor.

Genital tract

- In good light with a speculum and an assistant: perineum and episiotomy; vaginal walls and fornices; **all four quadrants of the cervix** (walk sponge forceps around it).
- Vulval or paravaginal swelling, severe pain, bluish tense mass = **haematoma**; rectal examination if a haematoma is suspected.
- A mass in the vagina or at the introitus = **inversion** (or a fibroid polyp, much rarer).

Placenta and membranes

Lay the placenta out: complete? Missing cotyledons, ragged membranes, vessels running off the edge to a torn edge (**succenturiate lobe**)?

Clotting

Bedside clotting test (WHO): 2 mL in a small dry glass tube held warm in your fist; tilt at 4 min, then every minute; **no clot by 7 minutes**, or a soft clot that breaks up, = coagulopathy. (Some units use the 20-minute whole-blood clotting test instead.) Oozing from drip sites also suggests coagulopathy.

Completing the examination

Thank her, cover her, hand hygiene, and **summarise** findings to the team with the measured loss, shock index, likely T(s) and the plan.

Document these positives	And these negatives
Measured loss (drape + weighed pads) and the time	"Placenta and membranes complete"
Fundal height and tone after massage	"Cervix inspected all round, no tear"
Shock index and urine output	"No haematoma, no inversion"
Site and degree of any tear	"Blood clots in the clotting test"

Red flags

RED FLAG

- Pulse >100–110, SBP <90–100 mmHg, **shock index ≥0.9** (≥1.4 = critical)
- Restlessness, confusion, air hunger, oliguria (<30 mL/h)
- Bleeding continuing after the first uterotonic, or a uterus that keeps relaxing
- **Shock out of proportion to visible bleeding**: think concealed bleeding (uterine rupture, broad-ligament or vulval haematoma, intra-abdominal bleeding after CS) or **uterine inversion**
- Blood that doesn't clot, oozing from cannula sites: DIC
- Severe perineal or rectal pain after delivery: expanding haematoma

RED FLAG

Call the most senior person available **immediately** if: shock index ≥1.4, SBP <90 mmHg, confusion or drowsiness, loss ≥1000 mL, blood that does not clot, or shock out of proportion to what you can see (think rupture, inversion, concealed haematoma, intra-abdominal bleeding after CS).

Thinking like a clinician

- **"Is the uterus hard?"** is the first branching question. Soft → Tone (massage, oxytocics, empty bladder, check for tissue). Hard but still bleeding → Trauma (inspect genital tract) or Thrombin (clotting test).
- **Bright red trickle with a well-contracted uterus** = a tear until proven otherwise, usually cervical or high vaginal. Clamp and suture; a second uterotonic won't fix it.
- **Uterus contracts then relaxes again and again** → think **retained tissue** (placenta incomplete) or overdistension/infection. Explore the uterus.
- **The fundus can't be felt, the woman is in profound shock (partly neurogenic, with bradycardia) and there is a mass in the vagina → uterine inversion**. Replace it immediately, before the cervix clamps down. **Stop the oxytocin until it is replaced.**
- **Previous CS + prolonged labour + sudden collapse** → uterine rupture: laparotomy, not another uterotonic.
- **After CS**: rising pulse, abdominal distension, falling Hb with little vaginal loss → intra-abdominal bleeding; return to theatre.
- **Anaemic woman**: her "500 mL" is someone else's 1500 mL. Treat the woman, not the number.
- **Common student mistakes**: underestimating loss by eye; waiting for hypotension; giving drugs one at a time; forgetting to catheterise; forgetting TXA; giving ergometrine to a hypertensive woman; forgetting to examine the placenta and the cervix; not documenting times.

EXAM FAVOURITE

Examiners love this: **"The uterus is well contracted but she is still bleeding. What will you do?"** Answer: inspect the genital tract under good light for trauma (cervical and vaginal tears, haematoma), check the placenta for completeness, do a bedside clotting test, and continue resuscitation.

7. Differential Diagnoses

These are differentials for the **cause** of bleeding (and for shock after delivery).

Differential	Distinguishing clinical feature	Distinguishing investigation
Uterine atony	Soft, boggy, large uterus above umbilicus; bleeding with clots; improves with massage	Palpation; response to massage/oxytocin
Genital-tract trauma	Uterus firm; continuous bright-red bleeding; instrumental or precipitate birth	Speculum examination in good light (± EUA)
Retained placenta / tissue	Placenta not delivered or incomplete; uterus relaxes repeatedly	Inspect placenta; manual exploration; ultrasound if available
Uterine rupture	Previous CS/obstructed labour; abdominal pain, shock > visible loss, fetal parts easily felt, haematuria	Clinical; laparotomy
Uterine inversion	Fundus not palpable abdominally; mass at introitus/in vagina; severe pain; shock with bradycardia	Clinical; immediate replacement
Coagulopathy / DIC	Non-clotting blood, oozing from sites; abruption, IUFD, AFE, sepsis, HELLP	Bedside clotting test, platelets, PT/APTT, fibrinogen

Differential	Distinguishing clinical feature	Distinguishing investigation
Vulval/vaginal/broad-ligament haematoma	Severe pain, rectal pressure, unexplained tachycardia, swelling; little external blood	Examination; ultrasound; drainage
Amniotic fluid embolism	Sudden collapse, hypoxia, hypotension, seizure, then DIC	Clinical diagnosis; ICU
Other shock (sepsis, PE, cardiac)	Fever, chest pain, dyspnoea; uterus fine and little bleeding	Clinical picture; ECG, etc.

8. Clinical Workup

In PPH the workup and the treatment happen **at the same time**. You never finish the history before you touch the uterus. Three principles:

1. **Send bloods while you resuscitate**; never delay treatment for a result. The first cannula gives you Hb, group and cross-match and the clotting-test tube.
2. **Bedside tests beat the lab at night**: the calibrated drape, weighed pads, shock index, HemoCue, the bedside clotting test and hourly urine output can all be done at a HC IV.
3. **Repeat, don't rely on one value**: the first Hb underestimates loss, and a normal first clotting test can become abnormal after a further litre of bleeding. Repeat Hb after resuscitation and at 24 h, and repeat clotting tests if loss exceeds about 1000 mL or bleeding continues.

Bedside

Test	How / why	Expected finding and what it changes	Available at
Calibrated drape reading	Read the level every 5–15 min; note the time	Objective loss; **the trigger for the bundle** (≥500 mL, or ≥300 mL with abnormal vitals)	HC III and above (where supplied)
Weigh pads, linen and swabs	Weigh soaked items and subtract the weight of the same items dry; **1 g ≈ 1 mL**	Adds the loss the drape missed (after the drape is removed, on the floor, in the bed)	All levels with a scale
Shock index	HR ÷ SBP, repeated with each set of vitals	Rising SI = ongoing bleeding even if BP is "normal"	All levels
HemoCue / point-of-care Hb	Finger-prick or venous sample	Baseline only; the **first Hb underestimates the loss**	HC III/IV and above
Bedside clotting test	2 mL venous blood in a clean, dry **glass** tube; hold it warm, tilt at 4 min then every minute (WHO MCPC); some units read at 20 min instead	No clot by 7 min, or a clot that breaks up = coagulopathy (DIC) → blood products, fresh whole blood	All levels
Hourly urine output	Foley catheter to a urometer or measured bag	<30 mL/h = under-resuscitated or AKI	All levels
Point-of-care ultrasound (if skilled)	Uterine cavity, abdomen	Retained products/clots in the cavity; free fluid (rupture, post-CS bleeding)	HC IV and above with a skilled operator
Urine dipstick for protein, BP	Undiagnosed pre-eclampsia?	Proteinuria + hypertension → avoid ergometrine; think HELLP	All levels
Malaria RDT, temperature	Secondary PPH, fever	Co-existing malaria, sepsis	All levels

Basic lab

Test	When / why	Expected / abnormal finding → management	Available at
Group and cross-match	Every PPH, with the first cannula	Blood group; 2–4 compatible units → transfuse early in major PPH	**HC IV**: only where there is a blood fridge and a supply from the regional blood bank (UBTS); otherwise refer. **RRH/NRH**: usually available, but stocks run out
Full blood count	Baseline, then after resuscitation and at 24 h	Hb (baseline often low from antenatal anaemia); platelets <75 × 10^9/L abnormal → transfusion and platelet decisions; Hb at 24 h guides iron versus transfusion	Most HC IVs and hospitals
Urea, creatinine, electrolytes	Major PPH, oliguria	Creatinine rising above ~80–90 µmol/L; high K^+ → AKI: fluid balance, avoid nephrotoxins (gentamicin dose), renal review	RRH/NRH
LFTs	Hypertensive woman, low platelets	Raised ALT/AST (with low platelets, haemolysis) → HELLP: coagulopathy, $MgSO_4$, no ergometrine	RRH/NRH

Imaging

Test	When / why	Expected finding → management	Available at
Ultrasound (formal)	Retained tissue suspected; secondary PPH; post-CS	Echogenic mass in the cavity; free fluid; haematoma → evacuation versus laparotomy versus expectant	Most HC IVs and all hospitals, if a sonographer is on site

Specialised / referral-level

Test	When / why	Expected / abnormal finding → management	Available at
PT/INR, APTT	Major PPH (≥1000 mL), abnormal clotting test, abruption, IUFD, sepsis, HELLP	Prolonged (>1.5 × normal) → FFP 15 mL/kg	**Mainly RRH/NRH/private hospitals**; often not at night
Fibrinogen	As above	**<2 g/L = severe coagulopathy** and predicts progression (normal in late pregnancy 4–6 g/L) → cryoprecipitate (or fresh whole blood if no components)	**NRH and some private hospitals only**
Lactate / blood gas	Severe shock, HDU/ICU	Lactate >4 mmol/L, base deficit → severity of shock; response to resuscitation	NRH/ICU only
Blood cultures, HVS	Secondary PPH or fever	Organisms and sensitivities → tailor antibiotics	RRH/NRH (culture capacity limited)
β-hCG	Secondary PPH that persists after evacuation	Raised or plateauing → suspect gestational trophoblastic neoplasia; send tissue for histology	HC IV (qualitative) to RRH/NRH (quantitative) **Verify** *quantitative β-hCG availability by level*

UGANDA CONTEXT

At most HC IVs you will have the **drape, a scale, a HemoCue, a glass tube and your hands**, and often no coagulation lab. That is enough to diagnose PPH, grade shock, find the cause (4 Ts) and detect coagulopathy. What you often **cannot** do is transfuse components or operate at night, so the workup must end in a clear decision: **manage here or refer now** (with NASG, tamponade, oxytocin running and an escort).

Putting it together

Model one-line summary

> "A 32-year-old para 5, 40 minutes after SVD of a 4.1 kg baby following an augmented labour, with primary PPH of 900 mL (drape plus pads), shock index 1.2, a soft uterus that firms with massage, a complete placenta and a second-degree tear: atonic PPH with a minor trauma component, haemodynamically compensated."

Problem list

1. Primary PPH (900 mL and still being measured), cause mainly **Tone**, plus **Trauma** (tear).
2. Hypovolaemia (SI 1.2), on the bundle.
3. Pre-existing anaemia (antenatal Hb 9.5 g/dL) → low reserve.
4. Risk of coagulopathy (continue clotting tests if loss exceeds 1000 mL).

Working diagnosis: *primary atonic PPH with a second-degree perineal tear, compensated hypovolaemic shock.*

Model case presentation (ward round)

> "Mrs N, a 32-year-old para 5 at 39 weeks, delivered a 4.1 kg baby by SVD 40 minutes ago after a 16-hour oxytocin-augmented labour. She developed primary PPH from uterine atony, with measured loss of 900 mL on the calibrated drape. Her pulse was 118, BP 96/60, shock index 1.2. The E-MOTIVE bundle was started at 10:05: massage, oxytocin 10 IU IV plus infusion, TXA 1 g IV, 1 L Ringer's, catheterised, and genital tract inspected (second-degree tear repaired, placenta complete). The uterus is now firm, pulse 100, BP 110/70, urine output 40 mL/h. Hb 8.1 g/dL; 2 units cross-matched. Plan: continue the oxytocin infusion, 15-minute observations, transfuse 1 unit, repeat Hb at 24 h, oral iron, and document as severe PPH for review."

EXAM FAVOURITE

Worked example 1: shock index and blood-loss estimation. The drape reads **650 mL** when it is removed. After that, four soaked pads and a draw sheet weigh **620 g**; the same items dry weigh **120 g**. Extra loss = 620 − 120 = **500 g ≈ 500 mL**. **Total ≈ 1150 mL = severe PPH.** Pulse 126, BP 90/60 → **SI = 126 ÷ 90 = 1.4** → urgent: second cannula, blood now, senior at the bedside, prepare theatre. The eye estimate in the notes said "about 500 mL": a classic underestimate. Her admission BP was 120/80 with pulse 86: SBP has fallen by 30 mmHg and pulse has risen by 40. By the **Rule of 30** she has lost roughly **30% of her blood volume** (moderate shock), which fits the weighed total.

EXAM FAVOURITE

Worked example 2: the first Hb and the clotting test. Thirty minutes after a 1500 mL PPH following abruption with IUFD, her Hb is **9.8 g/dL** (antenatal 11.2). This does **not** mean the loss was small: she has lost whole blood and there has been no time for haemodilution, so the Hb falls further over hours (and with crystalloid). Treat the **clinical picture and the measured loss**, not the first Hb. The bedside clotting test shows **no clot at 7 minutes**. Later labs: platelets **62 × 10^9/L**, PT prolonged, fibrinogen **1.3 g/L**. Interpretation: **DIC** from abruption plus consumption. Action: TXA (if within 3 h), red cells plus **FFP 15 mL/kg**, **cryoprecipitate** (fibrinogen <2 g/L), **platelets** (<75 × 10^9/L); fresh whole blood if components are not available; keep her warm; ensure the uterus is empty and contracted.

9. Management

Immediate / emergency

Diagnose early. Calibrated drape reading **≥500 mL**, or **≥300 mL plus abnormal vital signs or clinical concern** (WHO/FIGO/ICM 2025), or any bleeding with a woman who looks unwell, triggers the bundle. Start all components **together, within 15 minutes**.

First 5 minutes

1. **Recognise and name it.** Say out loud: "This is PPH. Time is 10:05."
2. **Call for help**: second midwife, the most senior clinician, anaesthetic officer; alert theatre and the blood bank. Assign roles (one on the uterus, one on IV and drugs, one on airway and observations, one scribe/timekeeper).
3. **Hand on the fundus at once.** Soft and high → start **massage** now. This is both the first examination and the first treatment.

ABCDE

- **A**: airway and conscious level.
- **B**: RR, SpO_2; **oxygen** 6–8 L/min by mask.
- **C**: pulse, BP, **shock index** (**≥0.9 abnormal, ≥1.4 urgent, ≥1.7 critical**). Lie flat, raise legs. **Two large-bore cannulae (14–16G)**; from the first draw blood for **Hb, group and cross-match**, and fill a **dry glass tube for the bedside clotting test**. Catheterise.
- **D**: AVPU; glucose if drowsy.
- **E**: temperature; **keep her warm**; look at the drape, the bed and the floor.

The E-MOTIVE bundle: emergency algorithm (vaginal birth, any facility)

1. **Call for help** and note the time (above).
2. **ABC**: lie flat, raise legs; oxygen; keep warm.
3. **Two large-bore IV cannulae**; bloods and clotting test.
4. **M: uterine Massage**. Rub the fundus firmly to stimulate a contraction and expel clots. Continue until firm.
5. **O: Oxytocin** 10 IU IV slowly (or IM), then an infusion (drug table).
6. **T: Tranexamic acid 1 g IV over 10 minutes** (within 3 hours of birth). Repeat 1 g after 30 minutes if bleeding continues.
7. **I: IV fluids**: warmed crystalloid (Ringer's lactate or normal saline). Give a 500 mL–1 L bolus quickly, then reassess. Avoid more than about 2 L of crystalloid before blood; start **blood** as soon as available.
8. **Empty the bladder** (Foley catheter, left in to measure urine output hourly).
9. **V: Vaginal/genital-tract Examination**. Is the placenta out and complete? Look for tears (repair them), haematoma or inversion.

10. **E: Escalate** if bleeding continues despite the bundle. Move to second-line drugs, bimanual compression, tamponade, theatre, and **refer early** from a HC III/IV that cannot operate or transfuse.
11. **Monitor** pulse, BP, RR, SpO_2 and blood loss every 15 minutes; urine hourly; **document** drug times and volumes.

> NOTE
> Why a bundle? In the **E-MOTIVE trial** (210,132 vaginal births, 78 hospitals in Kenya, Nigeria, South Africa and Tanzania), the calibrated drape plus the bundle **reduced the composite of severe PPH (≥1000 mL), laparotomy for bleeding or maternal death from bleeding from 4.3% to 1.6% (RR 0.40)**. Treatment was started early and together instead of sequentially after each "wait and see". WHO adopted it in 2023.

Definitive

Escalation: bleeding continues after the bundle

Work through these **in parallel**, not one by one, while someone prepares theatre and blood.

1. Bimanual uterine compression (buys time)

- One gloved fist in the anterior vaginal fornix pushes against the anterior uterine wall. The other hand, on the abdomen, presses the fundus down and forward onto the fist. The uterus is compressed between the two hands.
- Maintain firm pressure for at least 5–10 minutes (up to 20–30), until bleeding stops or help arrives.
- **External aortic compression** (fist pressed just above and left of the umbilicus onto the aorta until the femoral pulse disappears) is a temporising alternative, especially during transfer.

2. Uterine balloon tamponade (UBT): for **atonic** PPH not responding to uterotonics.

- The WHO (2021) recommends UBT **where standard PPH care, including surgery and blood, is accessible**. In practice in Uganda the **condom catheter** is widely used as a bridge to theatre or during referral.
- **Condom catheter**: tie a condom over a Foley catheter (or use a purpose-made balloon). Insert it into the uterine cavity past the cervix. Fill with **300–500 mL** warm saline through a giving set until bleeding stops, and pack the vagina with gauze to hold it in. Continue the **oxytocin infusion**, give **prophylactic antibiotics** (e.g. ampicillin 2 g IV single dose, or a cephalosporin), catheterise the bladder, monitor closely. Deflate gradually and remove after **6–24 hours** (usually 12–24 h), with theatre and blood ready.
- **Tamponade test**: if bleeding stops, the tamponade is working. If bleeding continues through or around the balloon, go straight to laparotomy.

3. Non-pneumatic anti-shock garment (NASG): neoprene segments wrapped from the ankles upwards to the abdomen. They compress the lower body and pelvis, squeezing blood back to the heart, brain and lungs and reducing pelvic bleeding. It is a **first-aid device to buy time**, particularly for **transfer from HC III/IV to hospital**. Apply ankle segments first; remove only when stable (SBP ≥90 and pulse <100, blood running), **from the ankles upwards**, one segment at a time, rechecking BP after each (a drop of 20 mmHg or a pulse rise of 20 means reapply).

4. Transfuse: whole blood or packed red cells as soon as possible. In major haemorrhage give **FFP** (15 mL/kg) when more than about 4 units of red cells have been given or clotting is abnormal, **cryoprecipitate** if fibrinogen is <2 g/L, and **platelets** if <75 × 10^9/L (availability at RRH/NRH). Aim for Hb >8 g/dL once bleeding is controlled.

5. Surgery (laparotomy, at CEmONC level), from the most conservative step upward:

Step	What / how	Why
Exploration	Evacuate clots, check for rupture, repair lacerations	Finds the trauma you missed

Step	What / how	Why
B-Lynch compression suture	Absorbable suture (e.g. No. 1 or 2 chromic/polyglactin on a large needle) that "braces" the uterus over the fundus; test first with bimanual compression (if that stops bleeding, B-Lynch will too)	Mechanically compresses an atonic uterus while preserving fertility
Bilateral uterine artery ligation (O'Leary stitch)	Ligate the ascending branches at the level of the lower segment, including some myometrium	The uterine arteries carry about 90% of uterine blood; cuts pulse pressure
Utero-ovarian / internal iliac ligation	Specialist procedures	Further devascularisation
Subtotal (or total) hysterectomy	**Decide early**; don't wait until she is in DIC	Life-saving. Subtotal is faster and safer in an unstable woman; total is needed if the lower segment or cervix bleeds (praevia/PAS, cervical tears)

EXAM FAVOURITE

"When would you do a hysterectomy?" Answer: when **conservative measures have failed** and the mother's life is at risk, **before** she becomes moribund. A living mother without a uterus is better than a dead mother with one. Get consent early (from the woman or her next of kin) and involve a senior.

Cause-specific management

Trauma

- **Perineal/vaginal tears**: repair in good light with analgesia (lidocaine 1%). Secure the **apex first** (start the stitch 1 cm above it), because retracted vessels keep bleeding.
- **Cervical tears**: sponge forceps on either side, suture from the apex. Refer if you cannot see the apex.
- **Vulval/vaginal haematoma**: small (<5 cm) and not expanding → ice, analgesia, observe. Large or expanding → incise, evacuate, ligate bleeders, pack; catheterise.
- **Uterine rupture**: resuscitate, then **laparotomy**: repair (if clean tear and she wants more children) or hysterectomy; inspect the bladder.
- **Uterine inversion**: call for help, **stop oxytocin**, treat shock, and **replace immediately**. **Johnson manoeuvre**: push the fundus back through the cervix with the palm, the part that came out last going in first; don't remove the placenta first if still attached. If that fails, use **O'Sullivan hydrostatic** replacement (warm saline into the vagina with the introitus sealed by hand or a ventouse cup), or surgical correction (Huntington, Haultain). **After replacement, give oxytocin** and hold the uterus in place until it contracts. Give antibiotics.

Tissue: retained placenta

1. If the placenta has not delivered **30 minutes** after birth: empty the bladder, **oxytocin 10 IU IV/IM** (if not already given as part of AMTSL, or a repeat), and repeat **CCT**. **Don't give ergometrine** (tetanic contraction can trap the placenta).
2. If still retained, or if bleeding: **manual removal of placenta (MRP)** under analgesia or anaesthesia (ketamine, spinal or pethidine + diazepam). Give **single-dose prophylactic antibiotics** (e.g. ampicillin 2 g IV, or cefazolin 2 g IV). The hand follows the cord into the uterus and finds the cleavage plane with the ulnar edge of the fingers while the other hand steadies the fundus. Remove it whole, check it, re-explore, and follow with an oxytocin infusion.
3. If there is **no cleavage plane**, suspect **placenta accreta spectrum**. Stop, don't pull pieces out, and call a senior. Laparotomy or hysterectomy may be needed.

4. **Retained fragments**: manual exploration or **MVA / blunt curettage** with a large curette (the soft postpartum uterus perforates easily).

Thrombin (coagulopathy)

- Treat the cause (deliver an abruption, treat sepsis), give **TXA**, fresh whole blood (contains clotting factors and platelets, useful where components are unavailable), FFP, cryoprecipitate and platelets. Keep her **warm**; correct calcium and acidosis.

Drug table

> **DRUG DOSING**
> Confirm the dose on your unit protocol. These are the WHO / Uganda regimens.

Drug	Dose (adult)	Route	Frequency	Duration / max	Key cautions
Oxytocin (treatment, first line)	**10 IU**, then infusion **20 IU in 1 L** NS/RL at **60 drops/min**; maintenance 20 IU in 1 L at **40 drops/min**	IV slowly or IM; infusion IV	Bolus once, then continuous	Continue for at least 2–6 h after bleeding stops. **No more than 3 L of oxytocin-containing fluid** (water intoxication)	Rapid IV bolus → hypotension. Very safe otherwise. Some Ugandan units use 20–40 IU in 1 L; follow your unit
Misoprostol (treatment)	**800 µg** (4 × 200 µg)	**Sublingual**	Single dose	Once	Use if oxytocin unavailable or as an add-on. Shivering and fever (up to 40 °C) are common and harmless; don't mistake them for sepsis. Slower onset than oxytocin
Ergometrine (ergonovine)	**0.2 mg**	IM, or IV **slowly**	Repeat 0.2 mg IM after 15 min, then 0.2 mg every 4 h	**Max 5 doses (1 mg) in 24 h**	**Contraindicated in hypertension, pre-eclampsia, heart disease**, retained placenta (and an undelivered second twin). Causes vomiting and severe hypertension; can cause stroke or eclampsia
Carboprost (15-methyl PGF2α)	**0.25 mg**	**Deep IM** (or intramyometrial by specialist)	Every 15 min	**Max 8 doses (2 mg)**	**Contraindicated in asthma** (bronchospasm). Diarrhoea, vomiting, fever. Expensive and rarely stocked in Uganda
Tranexamic acid	**1 g** (10 mL of 100 mg/mL)	IV over **10 min** (1 mL/min)	Second 1 g if bleeding continues after 30 min or restarts within 24 h	Give **within 3 h of birth**	For **all causes** of PPH (tone, trauma, tissue). Too-fast injection causes hypotension
Oxytocin (AMTSL, first choice)	**10 IU**	IM or slow IV	Once, within 1 min of birth	Single dose	Store 2–8 °C; loses potency in heat

Drug	Dose (adult)	Route	Frequency	Duration / max	Key cautions
Heat-stable carbetocin	**100 µg**	IM or IV	Single dose	Once	Non-inferior to oxytocin (CHAMPION trial, which included Uganda). Use where oxytocin quality can't be assured. **For prevention only, not treatment**
Misoprostol (prevention)	**400–600 µg** (Uganda community: 600 µg = 3 × 200 µg)	Oral	Once, immediately after the baby is born	Once	Where injectables are not available (home births, community distribution); never before the birth
Ergometrine / oxytocin-ergometrine (prevention)	0.2 mg IM / 5 IU + 0.5 mg IM	IM	Once	Once	Only if no hypertension; more side effects
Tranexamic acid at CS	1 g IV in some units for high-risk women	IV	Once	Once	Not routine for all births; follow local protocol
Ampicillin (MRP, tamponade prophylaxis)	**2 g**	IV	Single dose	Once	Or cefazolin 2 g IV, or a cephalosporin
Ampicillin + gentamicin + metronidazole (secondary PPH with infection)	Ampicillin **2 g** 6-hourly; gentamicin **5 mg/kg** once daily; metronidazole **500 mg** 8-hourly	IV	As stated	Until afebrile 48 h, then oral amoxicillin + metronidazole to complete 7 days	Or local regimen (e.g. ceftriaxone + metronidazole); adjust gentamicin in AKI
FFP	**15 mL/kg**	IV	After ~4 units red cells or abnormal clotting	As needed	RRH/NRH
Cryoprecipitate	Per local protocol Verify *unit dose not stated in source*	IV	If fibrinogen <2 g/L	As needed	NRH and some private hospitals
Platelets	Per local protocol	IV	If <75 × 10^9/L with active bleeding	As needed	RRH/NRH
Ferrous sulphate + folic acid	Ferrous sulphate 200 mg + folic acid	Oral	Daily Verify *frequency*	Until Hb restored	IV iron at referral level; transfuse if symptomatic or Hb <7 g/dL
Lidocaine 1%	Local infiltration	Local	Once	For repair	Analgesia for tear repair

CLINICAL PEARL

Carbetocin is for prevention only. It is not a treatment drug for established PPH. And there is no point giving IV ergometrine to a woman whose BP is 160/110. If you remember nothing else about contraindications: **"Ergometrine: not for the hypertensive heart; carboprost: not for the wheezy chest."**

Referral criteria

Level	What to do	Refer up when
HC II / community	Misoprostol 600 µg orally for prevention; uterine massage	Any bleeding after birth: refer urgently
HC III (BEmONC)	AMTSL; full E-MOTIVE bundle (massage, oxytocin, TXA, fluids, genital examination, repair tears); MRP; MVA for fragments; bimanual compression; condom tamponade and NASG	**Early**: as soon as bleeding is not controlled by the bundle, or she needs blood or theatre → HC IV/hospital
HC IV / General Hospital (CEmONC)	All of the above **plus** blood transfusion, theatre: laparotomy, B-Lynch, uterine artery ligation, hysterectomy	Needs blood components, HDU/ICU, PAS surgery → RRH
RRH / NRH (e.g. Mulago-Kawempe, Mengo)	Blood components (FFP, cryo, platelets), HDU/ICU, specialist surgery for PAS, interventional options where available	n/a

How to refer: IV oxytocin running, NASG on and/or tamponade in place, a health worker escort who can do bimanual compression, a referral note with times and doses, and a phone call ahead.

Monitoring and follow-up

- During the bundle: pulse, BP, RR, SpO_2 and blood loss **every 15 minutes**; urine hourly; document drug times and volumes.
- After the bleeding stops: continue the oxytocin infusion for several hours. Vital signs, fundus and loss every 15 min for 2 hours, then half-hourly and hourly; fluid balance chart; urine output.
- **Hb check at 24 h**; **treat anaemia** (ferrous sulphate 200 mg + folic acid; IV iron at referral level; transfusion if symptomatic or Hb <7 g/dL).
- **Thromboprophylaxis** once bleeding is controlled (massive PPH is a VTE risk).
- **Debrief** the woman and family; explain the risk next pregnancy (recurrence about 15%); document well and plan her next delivery at a CEmONC facility.
- Watch for **Sheehan's syndrome** (failure of lactation, amenorrhoea, fatigue after major PPH).
- **Near-miss audit or MPDSR** review if there was a death.

Prevention at every birth (AMTSL)

Active management of the third stage of labour (AMTSL), per WHO:

1. **Exclude a second baby** (palpate the abdomen).
2. **Oxytocin 10 IU IM** (or IV slowly) **within 1 minute of birth**. This is the **most important component**.
3. **Delayed cord clamping** (1–3 minutes) unless the baby needs resuscitation.
4. **Controlled cord traction (CCT)** with counter-traction above the pubis, only during a contraction, by a skilled attendant. It is optional where attendants are not skilled.
5. **Assess uterine tone** abdominally **every 15 minutes for 2 hours** (sustained continuous massage is no longer recommended if oxytocin was given, but tone checks are).
6. **Examine the placenta and membranes** for completeness, and the genital tract for tears.
7. **Measure blood loss objectively** (calibrated drape placed under the buttocks after birth, WHO 2023).

UGANDA CONTEXT

In Uganda, **oxytocin 10 IU IM** is standard AMTSL at every facility. **Misoprostol 600 µg orally** (3 × 200 µg tablets) has been distributed for women who may deliver at home, to be taken immediately after the baby is born, and never before. Heat-stable carbetocin has been introduced in some humanitarian settings. At caesarean, oxytocin is given IV **slowly** (a rapid bolus causes hypotension).

Secondary PPH (24 h to 12 weeks)

1. Assess and resuscitate as for primary PPH. Most secondary PPH is modest, but it can be massive.
2. **Investigations**: FBC, blood culture and high vaginal swab, pelvic ultrasound (retained products), **β-hCG if bleeding persists** (to exclude GTN), malaria test if febrile.
3. **Antibiotics** if infection is suspected: **ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly** until afebrile for 48 hours, then oral amoxicillin plus metronidazole to complete 7 days (or local regimen, e.g. ceftriaxone + metronidazole).
4. **Uterotonics**: oxytocin infusion, misoprostol 800 µg SL, or ergometrine if no hypertension.
5. **Evacuation of retained products** by an experienced person (MVA or gentle curettage), **after antibiotics have started** (the infected uterus perforates easily).
6. Persistent bleeding: surgical options as above.

10. Complications

Disease-related

Timing	Complication
Immediate	Hypovolaemic shock, death, DIC, cardiac arrest
Organ damage	**Acute kidney injury** (ATN, cortical necrosis), ARDS, **Sheehan's syndrome** (pituitary necrosis → failed lactation, amenorrhoea, hypothyroidism, adrenal insufficiency)
Later	Severe anaemia, fatigue, failure of breastfeeding, **VTE**, post-traumatic stress and depression

Treatment-related

Complication	Cause
Loss of fertility	Hysterectomy
Bladder/ureteric injury	Laparotomy, devascularisation, hysterectomy
Infection	MRP, tamponade, curettage
Dilutional coagulopathy; fluid overload, pulmonary oedema	Large volumes of crystalloid
Water intoxication	Oxytocin (>3 L of oxytocin-containing fluid)
Transfusion reactions	Blood products
Hypertensive crisis, stroke, eclampsia	Ergometrine in a hypertensive woman
Bronchospasm	Carboprost in asthma
Shivering, fever	Misoprostol (harmless)
Asherman's syndrome	Curettage
Uterine perforation	Curettage of the soft postpartum or infected uterus

11. Prognosis

- With early detection and the bundle, most PPH is controlled without surgery; E-MOTIVE cut the composite of severe PPH, laparotomy for bleeding or death from bleeding from 4.3% to 1.6%.
- Death follows delay: late recognition (eye estimates, waiting for hypotension), drugs given one at a time, late TXA, no blood, late referral.
- **Recurrence in the next pregnancy is about 15%**.
- After major PPH, watch for Sheehan's syndrome, persisting anaemia, VTE and psychological sequelae.

12. Prevention

- **Antenatal**: treat anaemia (iron + folic acid; Hb target ≥11 g/dL), identify high-risk women (previous PPH, praevia, grand multipara, twins), **birth preparedness** (deliver in a CEmONC facility, identify blood donors, transport money).
- **Intrapartum**: partograph to avoid prolonged labour; judicious augmentation.
- **Third stage**: **AMTSL at every birth** (section 9) with a calibrated drape and 15-minute tone checks for 2 hours.
- **Community**: misoprostol 600 µg orally for women who may deliver at home.
- **After PPH**: plan the next delivery at a CEmONC facility.

13. Special Populations

- **Pre-eclampsia / hypertension**: **no ergometrine or Syntometrine**; use oxytocin, then misoprostol 800 µg SL (carboprost if available and she is not asthmatic). Think HELLP and low platelets. $MgSO_4$ itself contributes to atony.
- **Asthma**: **no carboprost**.
- **Heart disease**: avoid ergometrine; give oxytocin slowly; careful fluids.
- **Anaemia** (Hb 8 g/dL): decompensates after "only" 400 mL; transfuse earlier.
- **After caesarean**: think intra-abdominal bleeding; return to theatre.
- **Rh-negative**: anti-D as per protocol.
- **Previous CS with praevia**: anticipate placenta accreta spectrum; deliver at a hospital with blood and a senior surgeon.

14. Key Points

High-yield points

- Primary PPH: ≥500 mL in 24 h, or ≥300 mL plus any abnormal vital sign (WHO/FIGO/ICM 2025); severe ≥1000 mL; secondary 24 h to 12 weeks.
- Measure, don't guess: calibrated drape and weighed pads; the eye underestimates by 30–50%.
- Shock index ≥0.9 abnormal, ≥1.4 urgent; hypotension is late.
- Start the whole E-MOTIVE bundle together: massage, oxytocin, TXA, IV fluids, genital-tract examination, escalate.
- TXA 1 g IV over 10 min within 3 h of birth: "1-10-3-30".
- Firm uterus still bleeding → trauma or thrombin, not another uterotonic.
- Ergometrine: not for the hypertensive heart; carboprost: not for the wheezy chest; carbetocin is for prevention only.

- Retained placenta at 30 min: empty bladder, oxytocin, CCT, then MRP with antibiotic cover; never ergometrine.
- Inversion: stop oxytocin, replace immediately, then oxytocin.
- Decide on hysterectomy early, before DIC.

Numbers & doses

Item	Value
Primary PPH	**≥500 mL within 24 h** of birth; or **≥300 mL + any abnormal vital sign** (WHO/FIGO/ICM 2025)
Severe / major PPH	**≥1000 mL** (RCOG: 1000–2000 moderate, >2000 severe)
Secondary PPH	**24 h to 12 weeks** after birth
Retained placenta	Not delivered **30 min** after birth (WHO; UK 60 min)
AMTSL uterotonic	**Oxytocin 10 IU IM** within **1 min** of birth; tone checks **every 15 min for 2 h**
Treatment oxytocin	**10 IU IV slowly/IM**, then **20 IU in 1 L at 60 drops/min**, maintenance **40 drops/min**; max **3 L** oxytocin-containing fluid
Tranexamic acid	**1 g IV over 10 min**, within **3 h**; repeat 1 g after **30 min** if still bleeding ("1-10-3-30")
Misoprostol	Prevention **600 µg orally** (400–600 µg); treatment **800 µg sublingual**
Ergometrine	**0.2 mg IM/slow IV**, repeat after 15 min then 4-hourly; **max 1 mg (5 doses)/24 h**
Carboprost	**0.25 mg deep IM** every 15 min; **max 2 mg (8 doses)**
Carbetocin (heat-stable)	**100 µg IM/IV**, prevention only
Shock index (HR ÷ SBP)	Normal 0.7–0.9; **≥0.9 abnormal; ≥1.4 urgent; ≥1.7 critical**
Pad weighing	**1 g ≈ 1 mL** (subtract dry weight)
Clotting test	2 mL in a dry glass tube; **no clot by 7 min** = coagulopathy (WHO)
Fibrinogen	Normal at term 4–6 g/L; **<2 g/L** = severe coagulopathy → cryoprecipitate
Platelets	Transfuse if **<75 × 10^9/L** in active bleeding
FFP	**15 mL/kg** after about 4 units of red cells or with abnormal clotting
Urine output target	**≥30 mL/h**
Crystalloid before blood	Avoid more than about **2 L**
Balloon tamponade	**300–500 mL** warm saline; remove after **6–24 h**
NASG removal	When **SBP ≥90 and pulse <100**, ankles first; reapply if SBP falls 20 or pulse rises 20
Recurrence next pregnancy	About **15%**

Exam traps

- **"Wait until the drape reaches 500 mL."** Wrong if vital signs are abnormal: at **≥300 mL plus an abnormal vital sign**, it is already PPH.
- **"BP is normal, so the loss is small."** Hypotension is late; look at the **pulse and shock index**.
- **"Give a second uterotonic"** for a **firm** uterus that keeps bleeding. Wrong: find the **tear** or check **clotting**.
- **"Give drugs one at a time and wait to see."** Wrong: the E-MOTIVE bundle is given **together**.
- **"TXA helps whenever it is given."** Wrong: only within **3 h** of birth.

- **Ergometrine for retained placenta** (traps it) or for a hypertensive woman (stroke, eclampsia). Give oxytocin.
- **"Hb 10 at 30 minutes, so no transfusion needed."** The first Hb underestimates acute loss.
- **Oxytocin before replacing an inverted uterus.** Replace first, then give oxytocin.
- **"Carbetocin for established PPH."** It is for prevention only.
- **"Couvelaire uterus = hysterectomy."** No; the uterus usually contracts.

Questions seniors ask

Viva questions

1. *Define PPH.* ≥500 mL within 24 h of birth; severe ≥1000 mL. WHO 2025: act at ≥300 mL with abnormal vital signs. Secondary: 24 h–12 weeks.
2. *What are the components of AMTSL?* Uterotonic within 1 min (oxytocin 10 IU IM), CCT, and uterine tone assessment (plus delayed cord clamping and examining the placenta).
3. *Name the components of the E-MOTIVE bundle.* Massage, Oxytocics, TXA, IV fluids, Examination, Escalation, triggered by objective measurement with a calibrated drape.
4. *Why is TXA time-critical?* Fibrinolysis peaks early; benefit is seen only within 3 h and falls by about 10% for every 15 minutes' delay.
5. *Contraindications to ergometrine?* Hypertension/pre-eclampsia, cardiac disease, retained placenta (and a second twin undelivered).
6. *What is a Couvelaire uterus, and is it an indication for hysterectomy?* Blood extravasated into the myometrium from abruption. **No**; the uterus usually contracts.
7. *How do you manage uterine inversion?* Call help, stop uterotonics, resuscitate, immediate manual replacement (Johnson), then oxytocin.

On the ward round

- **What is your first action when the midwife calls "PPH"?** Call for help and note the time, put a hand on the fundus and massage, and start the bundle together while assessing ABC and the shock index.
- **The uterus is firm but she is still bleeding. What now?** Inspect the genital tract in good light (cervix all round, vagina, perineum), check the placenta is complete, and do a bedside clotting test; continue resuscitation and give TXA if within 3 h.
- **How do you measure blood loss here?** Calibrated drape under the buttocks from birth, then weigh pads and linen (1 g ≈ 1 mL, minus dry weight); visual estimates are 30–50% too low.
- **Why give TXA even for atony?** Fibrinolysis rises quickly after birth for every cause of PPH; WOMAN showed about a 30% reduction in death from bleeding when given within 3 h.
- **Which uterotonic would you avoid in this woman with pre-eclampsia, and what would you use?** Avoid ergometrine (and Syntometrine); use oxytocin, then misoprostol 800 µg SL (carboprost if available and she is not asthmatic).
- **Her Hb is 9.8 g/dL after a 1.5 L bleed. Is she fine?** No. The first Hb has not equilibrated; repeat it after resuscitation and at 24 h, and transfuse on clinical grounds and measured loss.
- **When would you refer from a HC III, and how?** As soon as bleeding is not controlled by the bundle, or she needs blood or theatre: with IV oxytocin running, NASG and/or tamponade, a health worker who can do bimanual compression, a referral note, and a phone call ahead.
- **When would you do a hysterectomy?** When conservative measures fail and her life is at risk, decided early by a senior, before she is moribund in DIC.

Mnemonics

- **4 Ts**: **T**one, **T**rauma, **T**issue, **T**hrombin (70/20/10/1).

- **E-MOTIVE** (E is for early detection): **M**assage, **O**xytocics, **T**ranexamic acid, **I**V fluids, **V**aginal (genital-tract) examination, **E**scalation.
- **HAEMOSTASIS** (the classic RCOG escalation ladder): **H**elp and hands on the uterus, **A**ssess and resuscitate, **E**stablish aetiology / ecbolics, **M**assage, **O**xytocin infusion / prostaglandins, **S**hift to theatre (EUA, bimanual compression), **T**amponade (balloon), **A**pply compression sutures (B-Lynch), **S**ystematic pelvic devascularisation, **I**nterventional radiology, **S**ubtotal or total hysterectomy.
- **"Ergo: no BP, no heart; carbo: no wheeze."**
- **TXA "1-10-3-30"**: **1** g, over **10** minutes, within **3** hours, repeat after **30** minutes if still bleeding.

OSCE tips

- On the PPH drill station, **say "I will call for help" first**, state the time, assign roles, verbalise ABC, and use the E-MOTIVE sequence out loud.
- Demonstrate **bimanual compression** correctly (fist in the anterior fornix, other hand pressing the fundus down and forward).
- Always **check the placenta**, and say you will inspect the cervix all round in good light.
- Mention **documentation** (times, doses, measured loss) and **debrief**.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define primary PPH (classic and WHO/FIGO/ICM 2025).	Classic: blood loss **≥500 mL** from the genital tract within **24 h** of birth. WHO/FIGO/ICM 2025 adds: diagnose and act at **≥300 mL plus any abnormal vital sign**. Severe PPH ≥1000 mL.
2	Define secondary PPH and name its commonest cause.	Abnormal bleeding from **24 h to 12 weeks** after birth. Commonest cause: **endometritis**, often with **retained products**.
3	The 4 Ts of PPH with approximate proportions.	**Tone** ~70%, **Trauma** ~20%, **Tissue** ~10%, **Thrombin** <1%.
4	Components of active management of the third stage (AMTSL).	1. **Oxytocin 10 IU IM** within 1 min of birth (most important) 2. Controlled cord traction 3. Uterine tone assessment every 15 min for 2 h Plus delayed cord clamping (1–3 min) and checking the placenta.
5	What does E-MOTIVE stand for?	**E**arly detection with a calibrated drape, then the bundle: **M**assage, **O**xytocic drugs, **T**ranexamic acid, **I**V fluids, **V**aginal/genital-tract examination, **E**scalation.
6	Main result of the E-MOTIVE trial (NEJM 2023).	Severe PPH, laparotomy for bleeding or death from bleeding fell from **4.3% to 1.6%** (RR 0.40, i.e. a **60% reduction**) in 210,132 vaginal births in Kenya, Nigeria, South Africa and Tanzania.
7	Tranexamic acid for PPH: dose, route, timing, repeat.	**1 g IV over 10 min**, **within 3 h of birth**; repeat 1 g if bleeding continues after **30 min** or restarts within 24 h (WOMAN trial). Memory: **1-10-3-30**.
8	Oxytocin regimen for treating PPH (WHO).	**10 IU IV slowly/IM**, then **20 IU in 1 L** NS/RL at **60 drops/min**, then 20 IU in 1 L at **40 drops/min**. Max **3 L** of oxytocin-containing fluid (water intoxication).
9	Ergometrine: dose, maximum, contraindications.	**0.2 mg IM** (or slow IV); repeat after 15 min, then 4-hourly; **max 5 doses (1 mg)**. Contraindicated in **hypertension/pre-eclampsia, heart disease**, retained placenta.
10	Carboprost: dose, maximum, key contraindication.	**0.25 mg deep IM** every 15 min, **max 8 doses (2 mg)**. Contraindicated in **asthma** (bronchospasm).
11	Misoprostol doses: prevention vs treatment of PPH.	Prevention: **400–600 µg orally** (Uganda community distribution: 600 µg). Treatment: **800 µg sublingual**. Side effects: shivering, fever.
12	Heat-stable carbetocin: dose and role.	**100 µg IM/IV** single dose, for **prevention only** where oxytocin quality or cold chain can't be assured. Non-inferior to oxytocin (CHAMPION trial). **Not** for treatment.
13	Shock index: formula and thresholds in obstetrics.	**HR ÷ SBP**. Normal postpartum 0.7–0.9. **≥0.9** = concern; **≥1.4** = critical, needs urgent intervention or transfer. It rises before BP falls.
14	How do you perform bimanual uterine compression?	Fist in the **anterior vaginal fornix** pressing on the anterior wall; the other hand on the abdomen pushes the fundus down and forward onto the fist. Hold firm pressure until bleeding stops (at least 5–10 min).
15	Condom catheter tamponade: key steps.	Condom tied over a Foley catheter, inserted into the uterine cavity, filled with **300–500 mL** warm saline until bleeding stops; vaginal pack; continue **oxytocin infusion**; **antibiotic prophylaxis**; catheterise; remove after **6–24 h**. If bleeding continues, go to laparotomy.

#	Question	Answer
16	When is a placenta 'retained' and what is the first step?	Not delivered **30 min** after birth (WHO). Empty the bladder, give **oxytocin 10 IU IM/IV**, repeat CCT. If unsuccessful or bleeding → **manual removal** under anaesthesia with **single-dose prophylactic antibiotic**. Don't give ergometrine.
17	Uterine inversion: signs and immediate management.	Fundus not palpable abdominally, mass in the vagina, **shock out of proportion to blood loss** (often bradycardia). Call help, **stop oxytocin**, resuscitate, **immediate manual replacement (Johnson)**, then give oxytocin. Hydrostatic (O'Sullivan) if that fails.
18	Surgical options for refractory atonic PPH, in order.	Exploration → **B-Lynch** compression suture → **bilateral uterine artery ligation** → internal iliac ligation (specialist) → **subtotal/total hysterectomy**. Decide on hysterectomy early.
19	Bedside test for coagulopathy at a HC IV.	**20-minute whole blood clotting test**: 2 mL of venous blood in a clean dry glass tube; if no clot at 20 minutes, suspect coagulopathy (DIC).
20	Sheehan's syndrome: what, and how does it present?	Ischaemic necrosis of the enlarged pituitary after massive PPH. Presents with **failure of lactation**, amenorrhoea, fatigue, hypothyroidism, adrenal insufficiency (hyponatraemia, hypoglycaemia).
21	NASG: what it does and how to remove it.	Non-pneumatic anti-shock garment: compresses the legs and pelvis, raising central blood volume and reducing pelvic bleeding. A **first-aid bridge for transfer**. Remove only when stable (SBP ≥90, pulse <100), **starting at the ankles**, one segment at a time; reapply if BP falls by 20 mmHg or pulse rises by 20.
22	PPH: what must be done in the first 5 minutes?	Say 'PPH' and note the time; **call for help** and assign roles; **hand on the fundus and massage**; ABC, vitals and **shock index**; lie flat, oxygen, keep warm; **two 14–16G cannulae** with bloods (Hb, group and cross-match, **20-min clotting tube**); start the **bundle together** (oxytocin, TXA, fluids, catheter, genital-tract exam).
23	How do you add weighed pads to the drape reading?	Weigh the soaked pads/linen, **subtract the weight of the same items dry**, and count **1 g ≈ 1 mL**. Example: 620 g wet − 120 g dry = **500 mL** extra. Visual estimates underestimate loss by 30–50%.
24	Why is the first Hb after a PPH misleading?	She has lost **whole blood** and there has not been time for **haemodilution**, so the first Hb **underestimates** the loss. Treat the measured loss and the clinical picture; repeat Hb after resuscitation and at **24 h**.
25	Blood-product triggers in major PPH.	Red cells early; **FFP 15 mL/kg** after about 4 units of red cells or with abnormal clotting; **cryoprecipitate** if fibrinogen **<2 g/L**; **platelets** if **<75 × 10^9/L**. Fresh whole blood if components are unavailable. Keep her warm.
26	Pulse 126, BP 90/60 after birth. Shock index and meaning?	**SI = 126 ÷ 90 = 1.4** → needs **urgent intervention**: senior at the bedside, blood now, prepare theatre or refer. (≥0.9 abnormal; ≥1.7 critical.)

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 29-year-old para 4 delivers a 4.2 kg baby at a HC IV. Ten minutes after the placenta delivers complete, the calibrated drape reads 350 mL. Her pulse is 112/min and BP 98/60 mmHg (shock index 1.14). The uterus is soft. Which is the MOST appropriate action?

A Observe for 15 minutes and act if the drape reaches 500 mL
B Give ergometrine 0.5 mg IV and wait to see the response
C Diagnose PPH and immediately start the full E-MOTIVE bundle
D Refer to the regional referral hospital without treatment
E Give misoprostol 800 µg sublingually alone and recheck in 30 minutes

2. A 24-year-old primigravida delivers by vacuum extraction. The placenta is complete and the uterus is firm and well contracted at the umbilicus, but a steady stream of bright red blood continues. Pulse 104/min. What is the MOST likely cause of her bleeding?

A Genital tract trauma
B Uterine atony
C Retained placental tissue
D Disseminated intravascular coagulation
E Uterine inversion

3. A 34-year-old para 3 with severe pre-eclampsia (BP 162/108 mmHg on $MgSO_4$) develops atonic PPH after vaginal birth. She has received oxytocin 10 IU IV, an oxytocin infusion, TXA 1 g and uterine massage, but the uterus remains soft. Which additional uterotonic is MOST appropriate?

A Ergometrine 0.2 mg IV
B Oxytocin-ergometrine (Syntometrine) 1 ampoule IM
C Carboprost 0.25 mg IV bolus
D Misoprostol 800 µg sublingual
E Methylergometrine 0.2 mg IM

4. A woman delivered 4 hours ago at home and arrives at a HC III with ongoing PPH. She is pale, pulse 124/min. A student suggests tranexamic acid. Which statement is MOST accurate? Advanced

A TXA should be given because it helps whenever it is given
B TXA is not expected to help because it is more than 3 hours since birth; focus on uterotonics, fluids, the cause and referral
C TXA should be given as 1 g IM because IV access is difficult
D TXA is contraindicated in all postpartum women because of thrombosis
E TXA is only for trauma causes of PPH, not atony

5. Thirty minutes after a normal delivery at a HC III, the placenta of a para 2 has not delivered despite oxytocin 10 IU IM at birth and one attempt at controlled cord traction. There is minimal bleeding and she is stable. What is the BEST next step?

A Give ergometrine 0.5 mg IV to expel the placenta
B Inject oxytocin into the umbilical vein
C Pull harder on the cord with fundal pressure
D Wait another 2 hours before any action
E Empty the bladder, give a further 10 IU oxytocin and repeat CCT; if unsuccessful, manual removal with antibiotic cover

6. Immediately after a fundal-pressure delivery and vigorous cord traction, a para 1 collapses: pulse 52/min, BP 70/40 mmHg, with only about 400 mL of visible blood loss. The fundus cannot be palpated abdominally and a mass is felt in the vagina. What is the MOST important immediate action? Advanced

A Start an oxytocin infusion at 60 drops/min
B Attempt manual removal of the placenta
C Call for help, stop uterotonics, and immediately replace the uterus manually
D Give atropine and wait for the heart rate to recover
E Arrange a pelvic ultrasound scan

7. Which intervention is the MOST effective single measure to prevent PPH at a vaginal birth?

A Routine episiotomy
B Sustained uterine massage for 30 minutes in every woman
C Early cord clamping within 30 seconds
D Prophylactic uterotonic (oxytocin 10 IU IM/IV) within 1 minute of birth
E Routine tranexamic acid for all vaginal births

8. A 30-year-old para 6 at a regional referral hospital has atonic PPH continuing despite the full E-MOTIVE bundle, misoprostol 800 µg SL, and 20 minutes of bimanual compression. Measured loss is 1800 mL; she has received 2 units of blood; theatre is being prepared. What is the MOST appropriate temporising measure while theatre is prepared? Advanced

A Uterine balloon (condom catheter) tamponade
B Third dose of TXA
C Carboprost 0.25 mg IV
D Vaginal pack of gauze only
E Stop fluids to allow permissive hypotension for 2 hours

9. A 27-year-old has had a PPH of 1500 mL after delivery of a stillborn baby with a large retroplacental clot. Blood is oozing from her cannula sites and the vaginal blood does not clot. The HC IV has no laboratory at night. What is the BEST bedside test to confirm your suspicion?

A Kleihauer–Betke test
B Urine dipstick for protein
C Random blood sugar
D 20-minute whole blood clotting test
E Peripheral smear for malaria parasites

10. At a HC III 3 hours' drive from the nearest hospital, a woman has PPH controlled only while bimanual compression is maintained. The bundle has been given. What is the BEST plan for transfer?

A Send her by boda-boda with her relatives and a referral note

B Wait until the bleeding stops completely before referral

C Apply NASG and/or balloon tamponade, keep IV oxytocin running, and transfer by ambulance with a health worker after calling ahead

D Give ergometrine 0.2 mg IV every 15 minutes during transport regardless of BP

E Remove the IV line so it does not come out during transport

11. A 25-year-old known asthmatic has atonic PPH not responding to oxytocin and misoprostol. Which drug should be AVOIDED?

A Ergometrine

B Tranexamic acid

C Oxytocin infusion

D Carboprost (15-methyl PGF2α)

E Ringer's lactate

12. Six hours after an emergency caesarean section for obstructed labour, a 31-year-old has pulse 128/min, BP 88/56 mmHg and a distended, tender abdomen. Vaginal loss is minimal and the uterus feels firm. Hb has fallen from 10.5 to 6.8 g/dL. What is the MOST appropriate management? Advanced

A Increase the oxytocin infusion and observe

B Give misoprostol 800 µg rectally

C Insert a balloon tamponade

D Transfuse 2 units and discharge to the ward

E Resuscitate with blood and return to theatre for re-laparotomy

13. What is the main mechanism by which bleeding from the placental bed is controlled after delivery?

A Contraction of the interlacing myometrial fibres compressing the spiral arteries

B Vasoconstriction of the spiral arteries by their smooth muscle

C Platelet plug formation alone

D Reduction in cardiac output after delivery

E Increase in plasma fibrinolytic activity

14. A para 3 is readmitted on day 10 after SVD with heavy fresh vaginal bleeding, temperature 38.6 °C, foul-smelling lochia and a tender, bulky uterus. Pulse 108/min, BP 110/70 mmHg. What is the BEST initial management?

A Immediate curettage before antibiotics

B IV ampicillin, gentamicin and metronidazole plus uterotonics, ultrasound, and evacuation of any retained products once antibiotics have started

C Oral amoxicillin and discharge home

D Emergency hysterectomy

E Tranexamic acid alone

15. During laparotomy for atonic PPH in a 22-year-old para 1 who strongly wishes more children, bimanual compression of the uterus on the operating table stops the bleeding. What is the MOST appropriate next surgical step? Advanced

A Total abdominal hysterectomy
B Internal iliac artery ligation by the general surgeon
C Close the abdomen and give more oxytocin
D Subtotal hysterectomy
E B-Lynch uterine compression suture

Answers & explanations

1. C · Diagnose PPH and immediately start the full E-MOTIVE bundle

C is right: WHO/FIGO/ICM 2025 advise diagnosing PPH at **≥300 mL with abnormal vital signs**. Tachycardia and a shock index >0.9 qualify, so the whole bundle (massage, oxytocin, TXA, IV fluids, examination, escalation) should start now. A is wrong: waiting for 500 mL is exactly the delay the new threshold aims to prevent. B is wrong: ergometrine is second-line and 0.5 mg is the wrong dose (0.2 mg), and giving single drugs sequentially is less effective than the bundle. D is wrong: never refer without first-response treatment; the HC IV can give all of the bundle. E is wrong: misoprostol alone, with a 'wait and see' interval, is not the bundle; oxytocin is first line.

2. A · Genital tract trauma

A is right: a **firm uterus with continuing bright-red bleeding after an instrumental birth** = trauma (cervical or vaginal laceration) until proven otherwise; inspect under good light and repair. B is wrong: atony gives a soft, boggy uterus. C is wrong: the placenta is complete, and retained tissue usually causes a uterus that relaxes repeatedly. D is wrong: DIC is rare without a trigger (abruption, AFE, sepsis) and presents with non-clotting blood and oozing. E is wrong: in inversion the fundus is not palpable at the umbilicus.

3. D · Misoprostol 800 µg sublingual

D is right: **misoprostol 800 µg SL** is a safe add-on uterotonic in hypertension. A, B and E are wrong: ergometrine and its combinations and derivatives cause vasoconstriction and severe hypertension; they are **contraindicated in pre-eclampsia** (risk of stroke or eclampsia). C is wrong: carboprost is safe in hypertension but is given **deep IM (0.25 mg)**, never as an IV bolus, and it is contraindicated in asthma.

4. B · TXA is not expected to help because it is more than 3 hours since birth; focus on uterotonics, fluids, the cause and referral

B is right: in the WOMAN trial, TXA reduced death from bleeding only when given **within 3 hours**, and there was no benefit (and possible harm) later. Management should focus on massage, uterotonics, fluids, finding the cause (4 Ts), NASG and urgent referral. A is wrong: benefit is time-dependent and falls about 10% per 15 min of delay. C is wrong: the recommended route is **IV over 10 minutes**; get access (IO or cut-down if needed). D is wrong: WOMAN showed no increase in thromboembolic events. E is wrong: TXA is recommended for PPH of **any cause** when given within 3 h.

5. E · Empty the bladder, give a further 10 IU oxytocin and repeat CCT; if unsuccessful, manual removal with antibiotic cover

E is right: for a retained placenta at 30 minutes, WHO recommends **additional oxytocin 10 IU** with CCT; a **full bladder** can prevent separation. If this fails, proceed to **manual removal** under anaesthesia with **single-dose prophylactic antibiotics**. A is wrong: ergometrine causes a tetanic contraction that can trap the placenta, and 0.5 mg is the wrong dose. B is wrong: WHO does not recommend umbilical vein oxytocin. C is wrong: excessive traction with fundal pressure risks cord avulsion and **uterine inversion**. D is wrong: prolonged waiting increases PPH and infection risk.

6. C · Call for help, stop uterotonics, and immediately replace the uterus manually

C is right: this is **acute uterine inversion**. Shock out of proportion to blood loss, with **vagal bradycardia**, and an absent fundus. Immediate manual replacement (Johnson manoeuvre) while the cervix is still lax is the priority, along with resuscitation; give oxytocin only **after** replacement. A is wrong: oxytocin before replacement contracts the cervical ring and traps the inverted fundus. B is wrong: removing the placenta before replacement increases bleeding. D is wrong: atropine doesn't correct the cause; delay allows the cervix to clamp down. E is wrong: it is a clinical diagnosis and imaging only delays treatment.

7. D · Prophylactic uterotonic (oxytocin 10 IU IM/IV) within 1 minute of birth

D is right: the **prophylactic uterotonic** is the most important component of AMTSL and reduces PPH by about 50–60%. A is wrong: routine episiotomy increases blood loss and trauma. B is wrong: sustained massage is not recommended when oxytocin is given; only tone checks every 15 min. C is wrong: **delayed** cord clamping (1–3 min) is recommended; early clamping doesn't prevent PPH and harms the baby's iron stores. E is wrong: TXA is not recommended routinely for prevention in all vaginal births.

8. A · Uterine balloon (condom catheter) tamponade

A is right: **uterine balloon tamponade** is recommended for refractory atonic PPH where surgery and blood are available, and it often stops bleeding or buys time; if bleeding continues through it, proceed straight to laparotomy. B is wrong: TXA is limited to a second dose, and further doses aren't recommended. C is wrong: carboprost is given IM or intramyometrially, not IV. D is wrong: a vaginal pack alone conceals ongoing intrauterine bleeding. E is wrong: prolonged permissive hypotension isn't appropriate; resuscitate with blood.

9. D · 20-minute whole blood clotting test

D is right: abruption with fetal death can cause **DIC**. The **20-minute whole blood clotting test** (no clot in a dry glass tube at 20 min) is a simple bedside test available anywhere. A is wrong: Kleihauer quantifies fetomaternal haemorrhage for anti-D dosing, not coagulopathy. B is wrong: proteinuria screens for pre-eclampsia (a risk factor for abruption) but doesn't show coagulopathy. C and E are wrong: useful in other contexts but irrelevant to the clotting question.

10. C · Apply NASG and/or balloon tamponade, keep IV oxytocin running, and transfer by ambulance with a health worker after calling ahead

C is right: the **NASG** and **tamponade** are temporising devices designed to keep a woman alive during transfer. Keep oxytocin running, send an escort who can do bimanual/aortic compression, and **call ahead** so blood and theatre are ready. A is wrong: this is unsafe transport with no skilled escort. B is wrong: waiting at a facility that can't transfuse or operate risks death. D is wrong: ergometrine is limited to 5 doses in 24 h and is contraindicated if hypertensive. E is wrong: IV access is essential.

11. D · Carboprost (15-methyl PGF2α)

D is right: **carboprost** (a prostaglandin F2α analogue) causes bronchoconstriction and is **contraindicated in asthma**. A is wrong: ergometrine is acceptable if she is normotensive without heart disease. B is wrong: TXA is safe and recommended. C and E are wrong: oxytocin and crystalloid are safe in asthma. Note that misoprostol (PGE1) does not cause bronchospasm.

12. E · Resuscitate with blood and return to theatre for re-laparotomy

E is right: **shock with a distended abdomen, falling Hb and little vaginal loss after CS** means intra-abdominal bleeding (a slipped uterine-angle ligature, extension of the incision into the broad ligament, bleeding vessels). Resuscitate and do a **re-laparotomy**. A, B and C are wrong: the uterus is contracted and the bleeding is not atonic or intrauterine, so uterotonics and tamponade won't stop it. D is wrong: transfusion without source control only delays the inevitable.

13. A · Contraction of the interlacing myometrial fibres compressing the spiral arteries

A is right: the **'living ligatures'**. Myometrial fibres contract and kink the vessels between them, which is why tone matters most. B is wrong: trophoblast invasion replaces the muscular wall of spiral arteries, so they **cannot** constrict. C is wrong: clotting is secondary and cannot control flow of 500–800 mL/min alone. D is wrong: cardiac output actually rises immediately after delivery (autotransfusion). E is wrong: fibrinolysis would **worsen** bleeding, which is why TXA helps.

14. B · IV ampicillin, gentamicin and metronidazole plus uterotonics, ultrasound, and evacuation of any retained products once antibiotics have started

B is right: this is **secondary PPH from endometritis ± retained products**. Resuscitate, give **broad-spectrum IV antibiotics** (ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily + metronidazole 500 mg 8-hourly) and uterotonics, scan, and evacuate retained products **after antibiotics have started**, by an experienced operator (a soft infected uterus perforates easily). A is wrong: curettage without antibiotic cover risks septicaemia and perforation. C is wrong: she is septic and bleeding, so needs admission. D is wrong: hysterectomy is a last resort. E is wrong: TXA alone doesn't treat infection or tissue.

15. E · B-Lynch uterine compression suture

E is right: if bimanual compression at laparotomy stops the bleeding (the 'compression test'), a **B-Lynch** brace suture is likely to work and preserves fertility. A and D are wrong: hysterectomy is for failure of conservative measures, or earlier if she is in extremis. It is not first choice in a young woman when a compression suture should work. B is wrong: internal iliac ligation is technically difficult, risks ureteric and venous injury, and is less effective than simpler steps. C is wrong: closing without definitive haemostasis risks re-bleeding and death.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Atonic PPH after a long labour at a HC IV

You are the intern on call at a HC IV in Mukono district at 02:00. A 33-year-old para 5 delivered a 4.0 kg baby 15 minutes ago after a 14-hour labour augmented with oxytocin. The midwife gave oxytocin 10 IU IM at birth and the placenta delivered by CCT. She now calls you: 'the drape is at 600 mL and still filling'.

1. What are your first actions in the first minute?

> **Call for help** (second midwife, anaesthetic officer, theatre team), note the time, and assign roles. Assess **ABC**: lie her flat, give **oxygen**, and insert **two large-bore cannulae (14–16G)**. Take blood for Hb, **group and cross-match** and a **bedside clotting test**. Start the **E-MOTIVE bundle** immediately; PPH is already diagnosed (≥500 mL).

2. What quick history and examination points will you check while resuscitating?

> History from the notes: parity, labour length, **oxytocin augmentation**, big baby (all **atony** risk factors); any **hypertension** (rules out ergometrine) or **asthma** (rules out carboprost); antenatal Hb, HIV status. Exam: pulse **120**, BP **100/64** (shock index 1.2), cool peripheries. The **fundus is soft and at 2 cm above the umbilicus** = atony. Check the **placenta**: complete. Bladder distended. Brief genital inspection: small first-degree tear, not bleeding much.

3. Outline the bundle you will give, with doses.

> - **Massage** the uterus and expel clots
> - **Oxytocin 10 IU IV slowly**, then **20 IU in 1 L RL at 60 drops/min**
> - **TXA 1 g IV over 10 min** (within 3 h of birth)
> - **IV fluids**: warmed RL 1 L fast, then reassess
> - **Catheterise** (empty bladder; hourly urine)
> - **Examine** the genital tract and placenta (done)
> - Plan **escalation** if bleeding continues

4. Fifteen minutes later the uterus keeps relaxing and loss is now 1100 mL; pulse 128. What next?

> **Escalate**: **misoprostol 800 µg SL** (or ergometrine 0.2 mg IM if normotensive), a **second TXA 1 g** if bleeding continues 30 min after the first, and **bimanual compression**. Re-explore the uterus for **retained clots or tissue**. Insert a **condom catheter balloon tamponade** (300–500 mL saline) if still bleeding. **Transfuse** as soon as blood is available (it is a HC IV, so request from the blood bank). Prepare theatre and get consent for laparotomy (B-Lynch, uterine artery ligation, hysterectomy). Consider an early call to the regional hospital if you lack blood or surgical capacity.

5. The tamponade stops the bleeding. How will you manage her over the next 24 hours?

> Continue the **oxytocin infusion** (20 IU in 1 L at 40 drops/min); give **prophylactic antibiotics** (e.g. ampicillin 2 g IV); vital signs every 15 min for 2 h, then half-hourly; **fluid balance** with urine output >30 mL/h; watch the drain of the balloon. Deflate and remove the balloon after 12–24 h in theatre-ready conditions. Check **Hb at 24 h**; transfuse if symptomatic or Hb <7 g/dL; oral iron and folic acid. Thromboprophylaxis once stable. Breastfeeding support (watch for Sheehan's). **Debrief** her and her husband, and document fully.

6. How would you present her on the morning ward round?

> 'Mrs K, a 33-year-old para 5, is day 0 after SVD of a 4 kg baby following a 14-h augmented labour, complicated by **severe atonic PPH of 1100 mL**. She required the E-MOTIVE bundle, a second-line uterotonic, bimanual compression and condom tamponade, and received 2 units of blood. She is now stable: pulse 92, BP 112/70, urine 50 mL/h, uterus firm, minimal loss. Hb 7.9. Plan: remove the tamponade today, continue the oxytocin infusion until then, repeat Hb, oral iron, and counsel her on family planning and on delivering in hospital next time.'

> **Take-away:** Measure, don't guess; give the bundle together and early; escalate in parallel (drugs, compression, tamponade, blood, theatre) instead of waiting to see whether each step works.

Case 2: A firm uterus but still bleeding

At Mengo Hospital, a 22-year-old primigravida has just had a vacuum delivery for prolonged second stage. The placenta delivered complete 10 minutes later after oxytocin 10 IU IM. The drape shows 700 mL and a steady trickle of bright red blood continues. Pulse 108, BP 112/72.

1. What is the single most useful examination finding to decide the cause?

> **Palpate the fundus.** If it is **firm, central and at or below the umbilicus**, atony is unlikely. Bright red continuous bleeding with a firm uterus after an instrumental birth suggests **trauma** (cervical, vaginal or perineal tear).

2. What differentials would you consider and how will you distinguish them?

> 1. **Cervical or high vaginal laceration**: speculum, good light, sponge forceps around the whole cervix.
> 2. **Perineal or episiotomy extension**: visible.
> 3. **Vulvovaginal haematoma**: severe pain, swelling, rectal pressure.
> 4. **Retained fragment**: placenta checked complete, less likely.
> 5. **Coagulopathy**: clotting test; check that blood forms clots.
> 6. **Uterine rupture** (less likely in a primigravida without a scar): abdominal pain, shock out of proportion.

3. Examination under good light shows a 3 cm left lateral cervical tear actively bleeding. How do you manage?

> Still start the bundle basics: IV access, bloods, **TXA 1 g IV** (it helps trauma bleeding too), fluids, catheter. Then **repair**: good light, assistant, analgesia or anaesthesia (move to theatre for EUA if the apex can't be seen). Grasp both sides with sponge forceps, place the **first suture above the apex**, and run continuous absorbable sutures (e.g. 2-0 polyglactin). Check haemostasis and the rest of the genital tract.

4. After repair, what monitoring and follow-up does she need?

> Observations every 15 min for 2 h, then hourly; fundus and loss checks; urine output. **Hb** check; iron and folate. Analgesia. Document the tear and repair. Explain to the woman what happened. Consider a **PPH case review** (why was the tear not seen earlier?).

> **Take-away:** A contracted uterus that still bleeds is not an oxytocin problem: look for trauma in good light, repair from above the apex, and remember that TXA also helps trauma-related PPH.

Case 3: Secondary PPH on day 9

A 28-year-old para 2 presents to a regional referral hospital 9 days after a home birth assisted by a traditional birth attendant. She reports heavy bleeding since morning with clots, fever and lower abdominal pain. Temperature 38.8 °C, pulse 116, BP 100/64. The lochia is foul-smelling.

1. What focused history will you take and why?

- **Place and conduct of birth**: TBA, prolonged labour, was the placenta complete? (retained products)
- **Onset of fever, rigors, pain**: endometritis
- **Urinary symptoms, breast pain, cough, calf pain**: other sources of fever
- **Malaria exposure**: common co-existing cause of fever in Uganda
- **HIV status/ART**: risk of severe sepsis
- Previous bleeding, contraception, breastfeeding
- **Danger symptoms**: dizziness, breathlessness, confusion

2. What examination findings and investigations do you want?

Exam: pallor, signs of sepsis (RR, BP, mental state), a **tender, bulky, soft uterus** above the expected size for day 9 (subinvolution), open os with tissue, foul lochia, perineal wound state, breasts, calves. Investigations: **FBC**, **blood cultures**, **HVS/endocervical swab**, **malaria RDT/smear**, **group and cross-match**, urea/creatinine, **pelvic ultrasound** for retained products, urinalysis. β-hCG if bleeding persists or tissue looks molar.

3. Outline management step by step.

1. Resuscitate: IV access, fluids, oxygen; transfuse if Hb low or unstable.
2. **IV antibiotics** immediately: **ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily + metronidazole 500 mg 8-hourly** (or ceftriaxone + metronidazole per local protocol).
3. **Uterotonic**: oxytocin infusion, or misoprostol 800 µg SL.
4. Treat malaria if positive.
5. Once antibiotics have started (usually within hours if bleeding is heavy): **gentle evacuation** of retained products (MVA or blunt curette) by a senior. Beware of perforation.
6. Continue antibiotics until afebrile for 48 h, then oral to complete 7 days.

4. Despite treatment she remains febrile on day 3 of antibiotics. What would you think of?

Consider **pelvic abscess** (ultrasound), **septic pelvic thrombophlebitis** (fever spikes despite antibiotics → heparin), infected haematoma, **wound infection**, **resistant organism** (review cultures, change to e.g. meropenem where available), **malaria**, UTI/pyelonephritis, mastitis, drug fever, and rarely retained products still present or GTN.

Take-away: Secondary PPH is usually infection plus retained products: give antibiotics first, then evacuate gently, and look for other causes of fever (including malaria) if she doesn't settle.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.